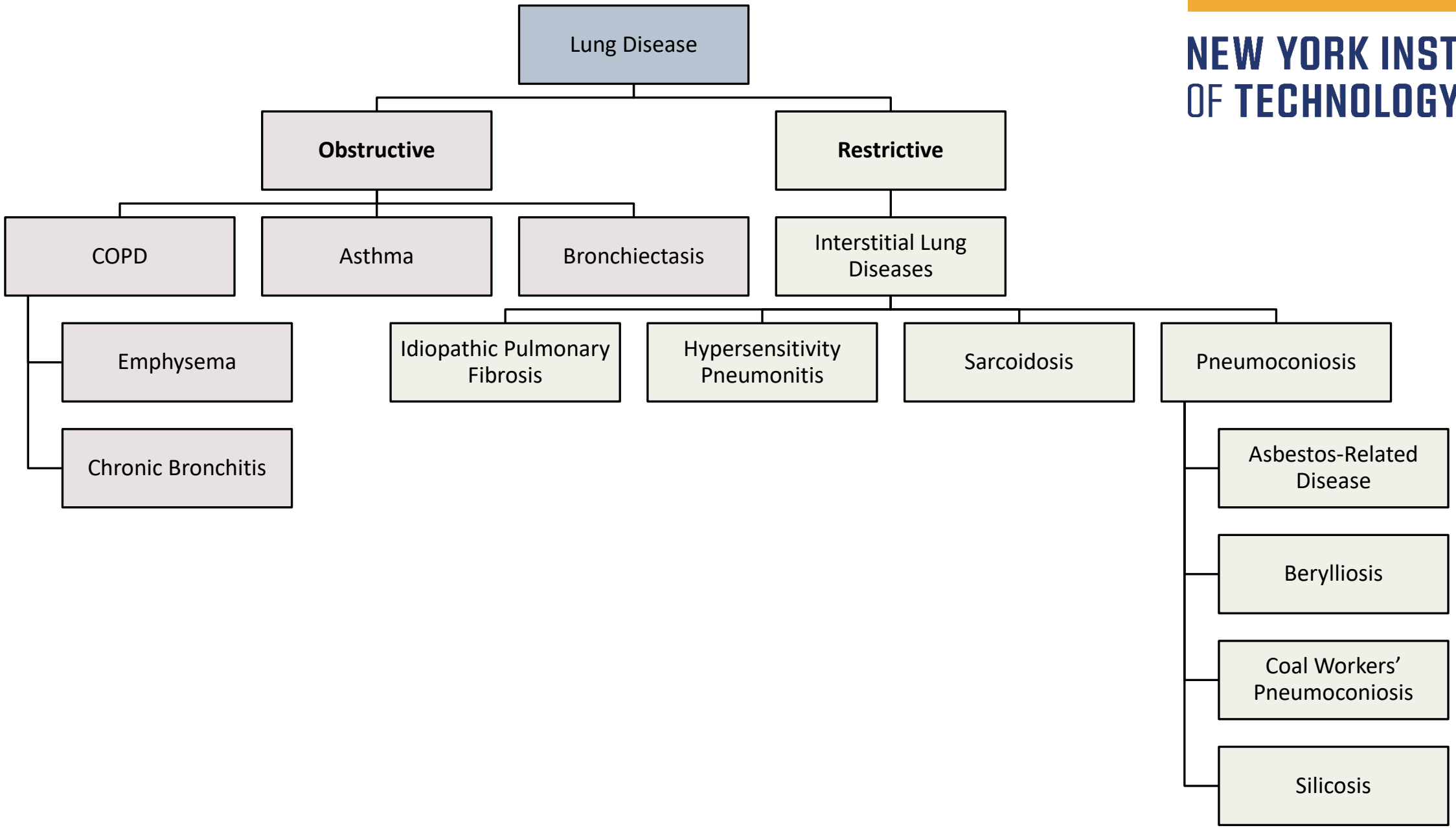
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Learning Objectives

1. Diagnose obstructive versus restrictive lung diseases based on spirometry and flow-volume loops
2. List symptomology and clinical presentations of obstructive and restrictive lung diseases
3. Describe the etiology and pathophysiology of obstructive and restrictive lung diseases
4. Identify diagnostic methodologies and findings of obstructive and restrictive lung diseases
5. Recognize risk factors for obstructive and restrictive lung diseases
6. Select appropriate treatment modalities including pharmacological and non-pharmacological management
7. Predict complications of obstructive and restrictive lung diseases based on evaluation of patient case scenarios

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Patient presents with long term shortness of breath and undergoes spirometry testing, which reveals **increased total lung capacity** and a decreased forced expiratory volume in one second (FEV1) and forced vital capacity (FVC) with **FEV1 is decreased more than FVC**.

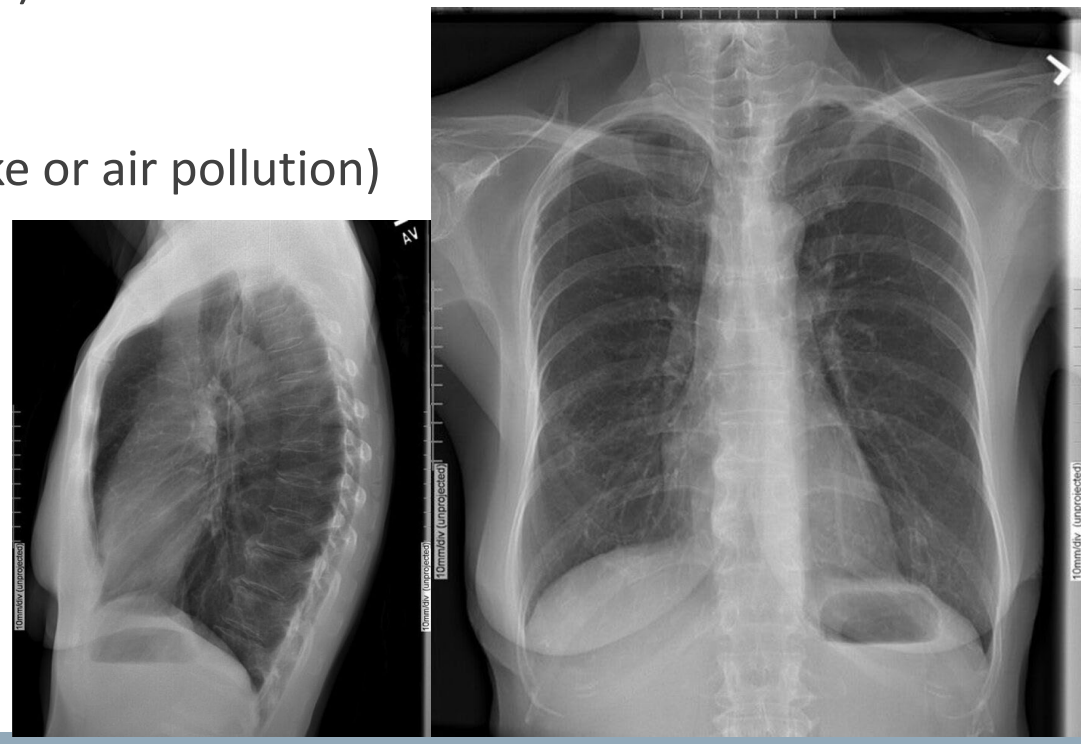
Is this patient presenting with obstructive or restrictive lung disease pattern?
Obstructive lung disease

What is on your differential for this classification of lung disease?
COPD including emphysema and chronic bronchitis, asthma, bronchiectasis

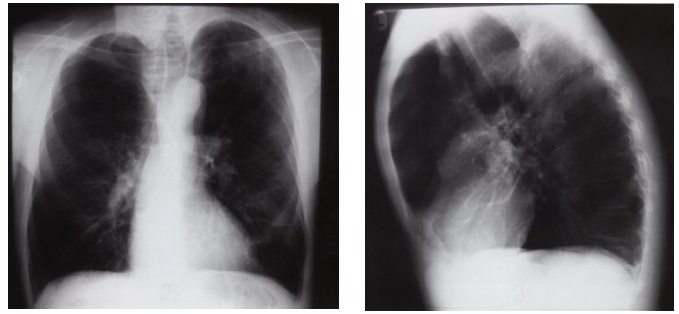
Chronic Obstructive Pulmonary Disease (COPD): Emphysema & Chronic Bronchitis

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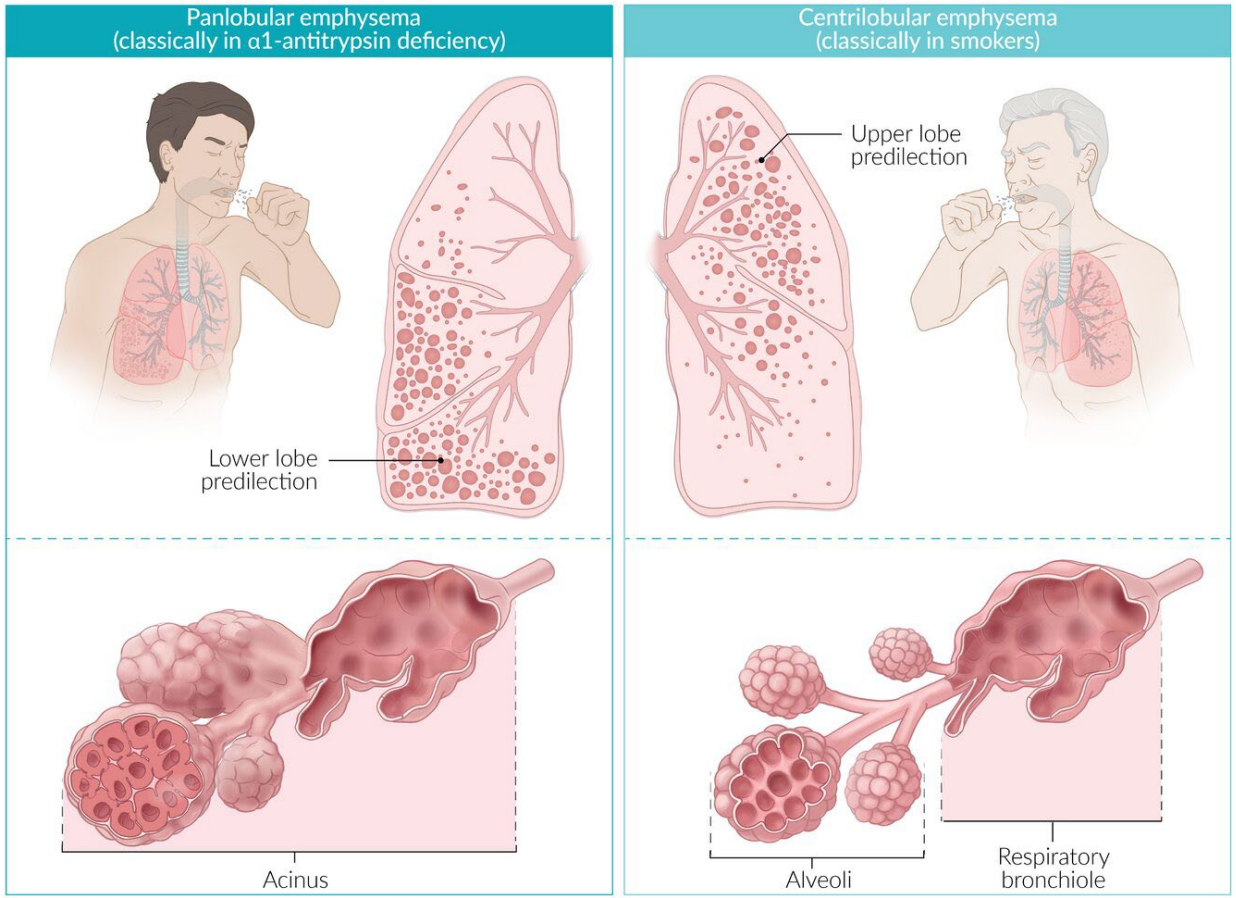
- Persistent respiratory symptoms (cough, dyspnea) resulting from airflow obstruction and alveolar gas exchange abnormalities
- Airflow limitation (postbronchodilator **FEV1:FVC ratio < 0.70**)
- Airway obstruction and parenchymal destruction
- Predominantly caused by inhaled toxins (e.g., tobacco smoke or air pollution)
- **CXR: pulmonary hyperinflation**
  - Increased AP diameter (barrel chest)
  - Flattened diaphragm
  - Widened intercostal spaces
  - Hyperlucency of lung tissue
  - Long and narrow heart shadow



Emphysema “Pink Puffer”



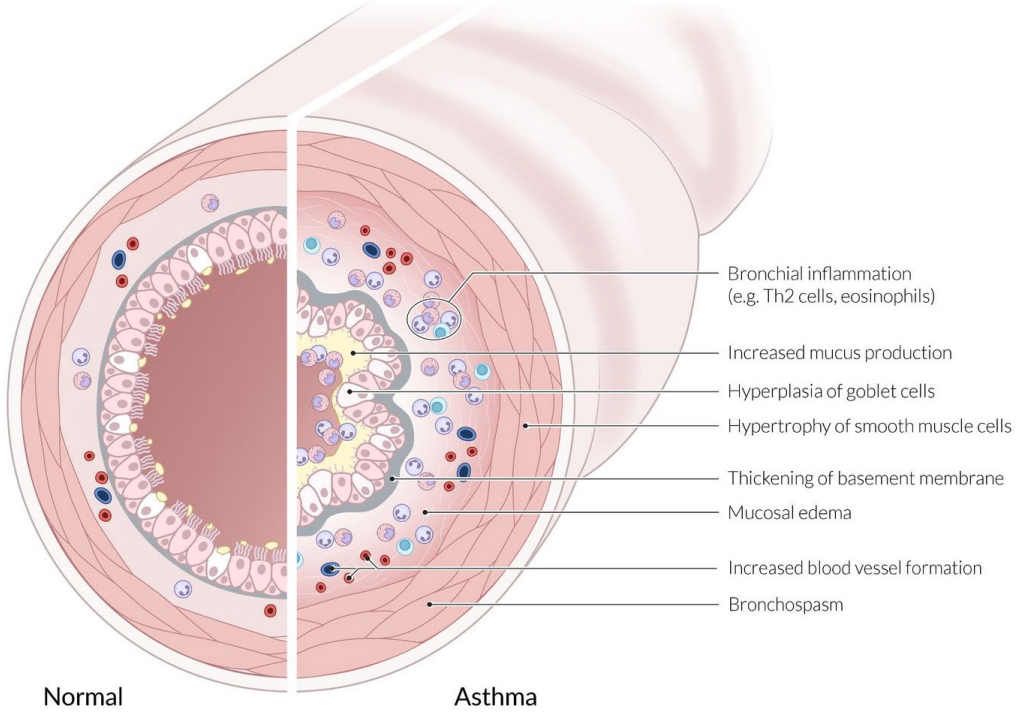
- Presents with **barrel-shaped chest** with prolonged expiration through pursed lips
- Pathophysiology:
  - Permanent **dilatation** of pulmonary air spaces distal to the terminal bronchioles that is caused by the destruction of the alveolar walls and pulmonary capillaries required for gas exchange
  - **Imbalance of proteases and antiproteases** → ↑ elastase activity → ↑ loss of elastic fibers → ↑ lung compliance, ↓ recoil, ↓ DLCO
- Panacinar
  - Associated with **alpha1-antitrypsin deficiency**
  - **Lower lobes**
- Centriacinar
  - Associated with **tobacco smoking**
  - **Upper Lobes**
- CXR: ↑ AP diameter, flattened diaphragm, hyperlucent lungs



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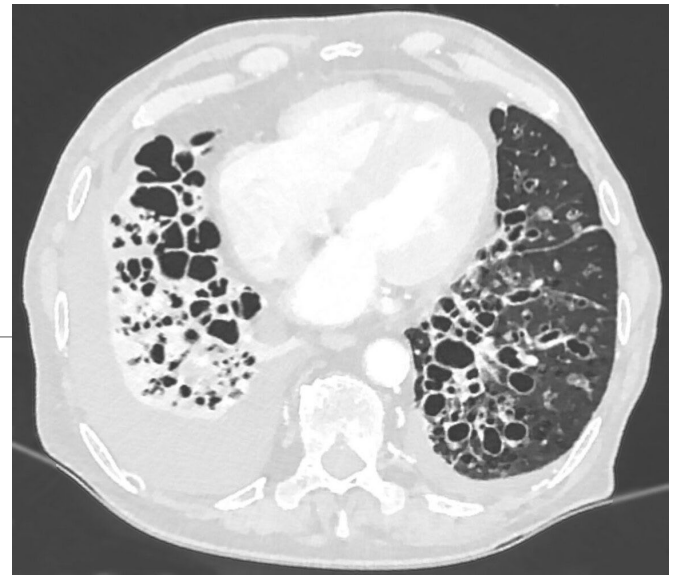
Asthma

- Unknown etiology with chronic airway inflammation driven by T-helper type 2 cells (Th2-cell)
- Pathophysiology: **Bronchial hyperresponsiveness, Bronchial inflammation, Endobronchial obstruction**
- Asymptomatic at baseline with episodes of persistent, dry cough, end-expiratory wheezing, tachypnea, dyspnea, or hypoxemia
- Allergic asthma
  - IgE-mediated type 1 hypersensitivity
  - Childhood
- Nonallergic asthma
  - Irritant or Aspirin-exacerbated respiratory disease (AERD)
  - > 40 years
- Risk factors: Atopy, Allergies, Genetic predisposition, Environmental and socioeconomic factors
- Triggers: viral URI, allergens, stress, aspirin
- Diagnosis
  - Prescence of typical asthma symptoms that vary in severity
  - Spirometry +/- bronchodilator responsiveness testing or bronchial challenge tests
  - Sputum analysis: Curschmann spirals and Charcot-Leyden crystals
- Complications: pulsus paradoxus, status asthmaticus



Bronchiectasis



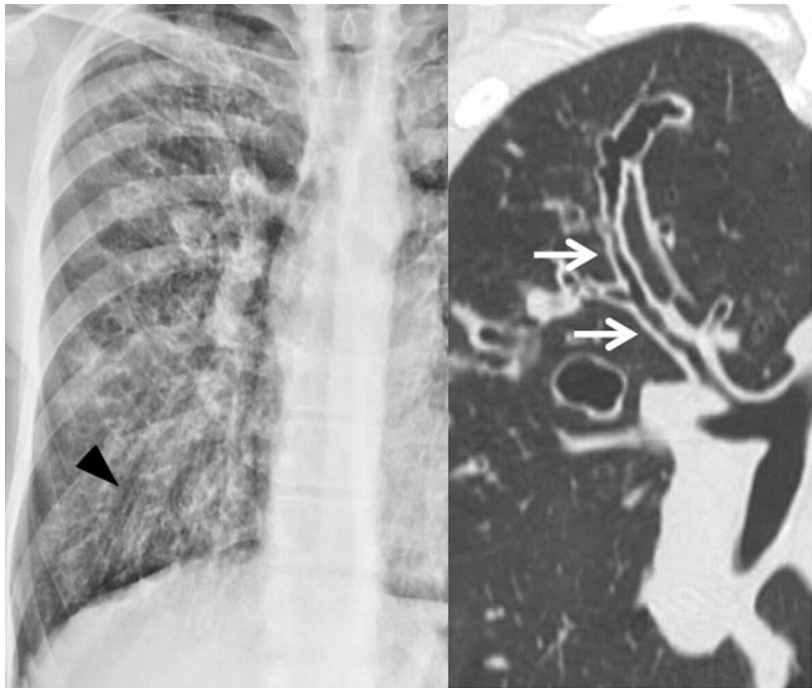


•Chronic necrotizing infection of bronchi or obstruction leads **to permanently dilated large airways** and thickened bronchial walls of small/medium airways

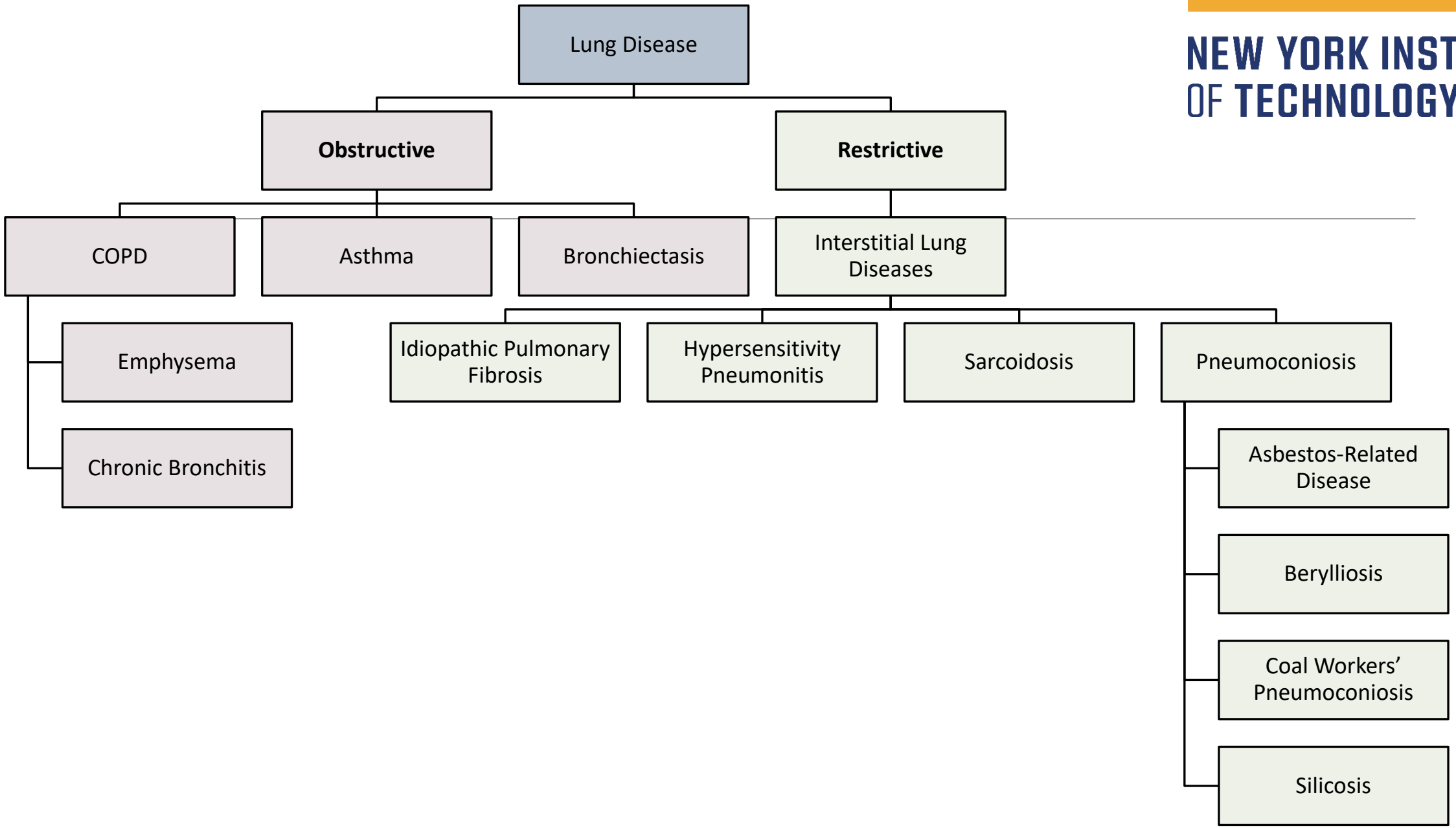
•Presents **with chronic (months-years), daily, copious, purulent sputum**, hemoptysis, digital clubbing, recurrent infections (*P. aeruginosa*)

•Associated with bronchial obstruction, poor ciliary motility (tobacco smoking, Kartagener syndrome), cystic fibrosis, allergic bronchopulmonary aspergillosis (ABPA), pulmonary infections (*M. avium*)

- Diagnosis: CXR (initial) and HRCT (confirmatory)
  - Bronchial dilation
  - Cylindrical or tubular (most common): parallel tram track sign and signet ring sign
  - Varicose
  - Saccular or cystic (most severe form)
  - Thickened bronchial walls, mucus plugging, honeycombing (suggests late-stage)
- Treatment
  - Acute
    - Supportive treatment and oxygen therapy,
    - Mucoactive agents and airway clearance techniques
    - Obtain a new sputum culture and start empiric antibiotic therapy (tailor to the most recent sputum culture)
  - Long term: stop/delay disease progression, reduce exacerbation frequency (goal ≤ 2 per year), achieve symptom control, and improve patient's QOL
- Complications:
  - Recurrent bronchopulmonary infections → chronic obstructive pulmonary disease → respiratory failure and cor pulmonale
  - Pulmonary hemorrhage (massive hemoptysis)
  - Lung abscess



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Patient presents with long term shortness of breath and undergoes spirometry testing, which reveals decreased total lung capacity and a normal forced expiratory volume in one second (FEV1) to forced vital capacity (FVC) ratio.

Is this patient presenting with obstructive or restrictive lung disease pattern?
Restrictive

What is on your differential for this classification of lung disease?
Interstitial Lung Diseases including Idiopathic Pulmonary Fibrosis, Hypersensitivity Pneumonitis, Sarcoidosis, Pneumoconiosis

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Idiopathic Pulmonary Fibrosis

- **Most common type of ILD**
- Irreversible pulmonary fibrosis and impaired pulmonary function with unknown etiology
- Associated with cigarette smoking, environmental pollutants, genetic defects
- Affects most commonly men 50–70 years of age
- Presents with progressive dyspnea, fatigue, nonproductive cough, crackles, clubbing
- Diagnosis of exclusion:
  - HRCT: **Peripheral reticular, ground-glass opacities with traction bronchiectasis +/- honeycomb appearance**
  - **Interstitial pneumonia (UIP)** pattern on histology
- Complications: pulmonary hypertension, RHF, arrhythmias, CAD, respiratory failure (usually within 3-7 years), lung cancer



Hypersensitivity Pneumonitis (extrinsic allergic alveolitis)

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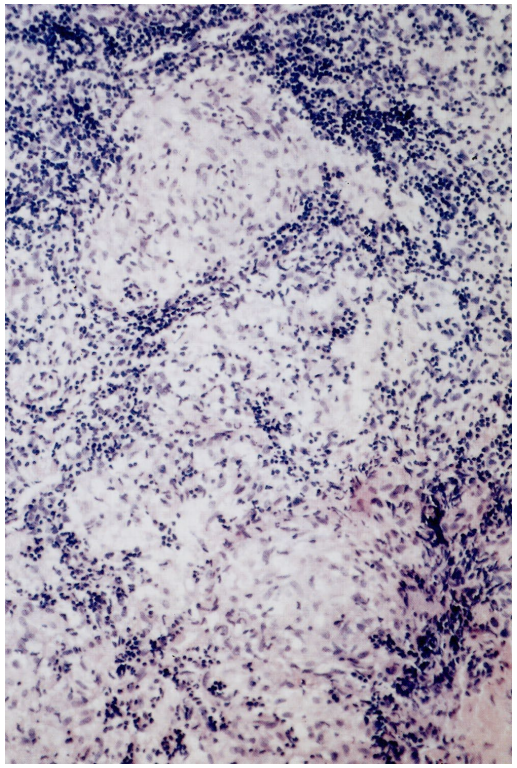
- Inhalation of **environmental and/or occupational antigens** (Farmer's lung, Pigeon breeder's lung, Chemical worker's lung)
- Combined **type III and type IV hypersensitivity reaction** with genetic predisposition
- Presents acutely with dyspnea, cough, chest tightness, fever, headache
- Diagnosis:
  - HRCT: **Honeycombing** (irreversible fibrotic changes), **thickening of alveolar septa**, **traction bronchiectasis or bronchiolectasis**, **ground-glass opacities** with inspiratory mosaic attenuation
  - Bronchoalveolar lavage (BAL): lymphocytosis (e.g., ≥ 20–30%)
  - Biopsy: **noncaseating granulomas** with lymphocytes and polynuclear giant cells
- Complications: Respiratory failure & Pulmonary heart disease
- Treatment:
  - Antigen avoidance & supportive management
  - Pharmacotherapy
    - Immunosuppressants (Glucocorticoid therapy, mycophenolate mofetil, azathioprine, rituximab)
    - Antifibrotic drugs (e.g., nintedanib, pirfenidone)



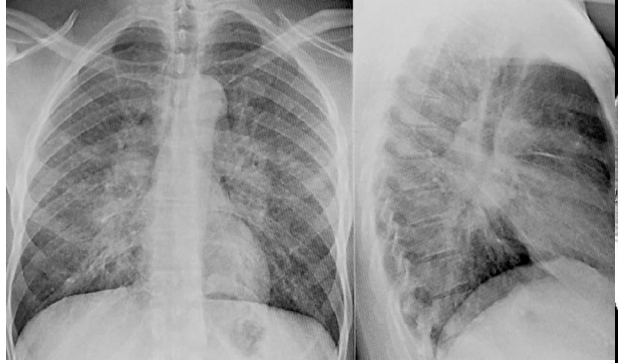
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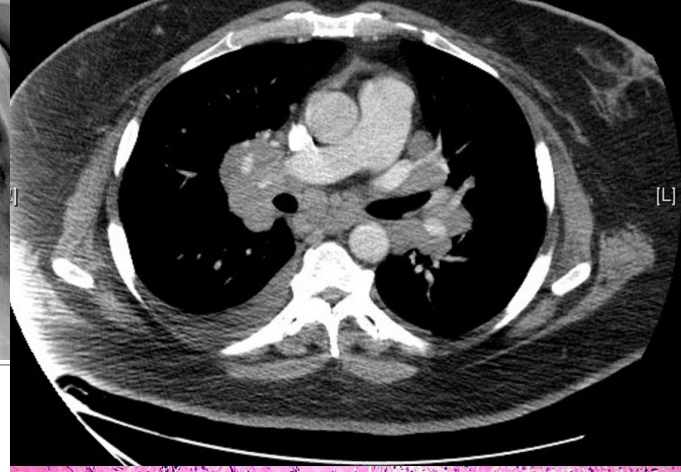
Sarcoidosis

- Unknown etiology but is thought to be related to an environmental exposure in a genetically predisposed individual
- Affects most commonly **African American or Scandinavian females 30–55 years of age**
- Symptomology:
  - Variable presentation from asymptomatic (early) to progressive, relapsing disease.
  - **Symptoms can include fever, malaise, lack of appetite, weight loss, lymphadenopathy, dyspnea, cough, chest pain, arthritis, erythema nodosum, anterior uveitis**
- Systemic disorder characterized by widespread, immune-mediated formation of **noncaseating granulomas**
  - Differentiation and clonal expansion of T cells/T cell recruitment and ↑ B-cell activity → local immune hyperactivity and inflammation → formation of noncaseating granulomas in the lungs and the lymphatic system



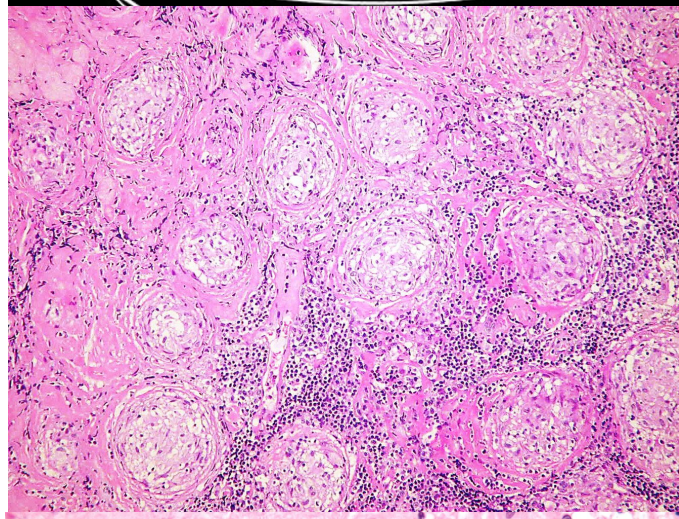
Sarcoidosis

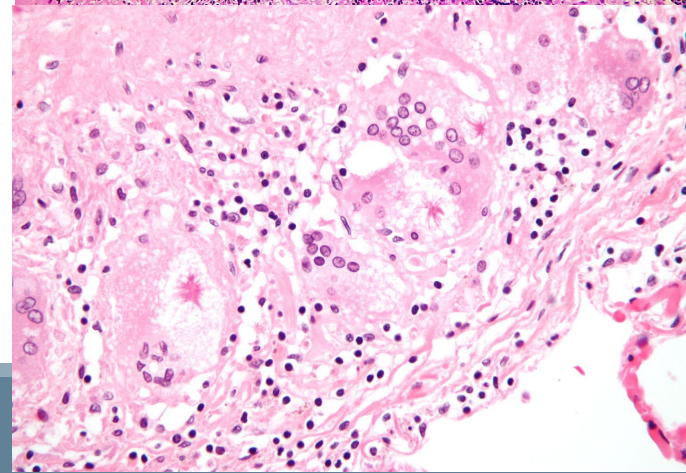




Diagnosis of exclusion:

- **Noncaseating granulomas** on biopsy
- Labs: leukocytopenia, ↑ESR, ↑CRP, **↑serum ACE levels, hypercalcemia, hypercalciuria,** ↑ IgG
- CXR: **bilateral hilar lymphadenopathy** with or without pulmonary infiltrates (reticular and/or ground glass opacities)
- HRCT:
  - **Extensive bilateral hilar and mediastinal lymphadenopathy**
  - Bronchocentric and perilymphatic micronodules (i.e., bronchovascular beading)
  - Peribronchial thickening
  - Fibrosis
- BAL:**↑ CD4:CD8 ratio**
- Biopsy [Gold standard]: Histopathology findings of **noncaseating granulomas with giant cells, Asteroid bodies, Schaumann bodies**





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Sarcoidosis

- Treatment
  - Supportive therapy: NSAIDs for pain relief, noninvasive ventilation, supplemental oxygen
  - Pharmacological therapy (immunosuppressive therapy) indicated for patients with symptomatic progressive disease, persistent pulmonary infiltration, or progressive decline of lung function
    - First line: **Glucocorticoids** (prednisone)
    - +/- DMARDs
      - Methotrexate or azathioprine
      - Chloroquine or hydroxychloroquine
      - Biologics (TNF-alpha inhibitors)
- Complications:
  - Malignancy (lung cancer, malignant lymphomas)
  - Bronchiectasis
  - Pulmonary hypertension
  - Aspergilloma
  - Chronic renal failure



Pneumoconiosis: Asbestos-Related Disease

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Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Asbestos-Related Disease	Airborne asbestos fibers	Asbestos miners and millers Brake linings and insulation manufactures Ship construction workers Demolishers Firefighters Plumbers Roofers	Symptoms typically develop 20–30 years after initial exposure Presents with exertional dyspnea, cough, digital clubbing Microscopic **asbestos bodies (ferruginous body)** in alveolar septa on histology Complications: Lung cancer (bronchogenic carcinoma > mesothelioma), Caplan syndrome	Diffuse bilateral infiltrates (**lower lobes**) Interstitial fibrosis **"Ivory white" calcified subdiaphragmatic and pleural plaques** (pathognomonic) Pleural effusion

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Pneumoconiosis: Silicosis

Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Silicosis	Crystalline silica dust Macrophages release fibrogenic factors in response to silica	Miners Workers of quarries, pottery production, sandblasting, glass manufacturers Construction workers	Acute silicosis: Dyspnea, cough, pleuritic pain Chronic silicosis: Chronic cough, Exertional dyspnea, Fatigue, Weight loss Complications: **Tuberculosis**, Progressive massive fibrosis, Respiratory failure, Cor pulmonale, Caplan syndrome	**Eggshell calcifications** of hilar lymph nodes Rounded, solitary, small opacities (**upper lobes**) Bilateral diffuse ground-glass opacities

Pneumoconiosis: Coal Workers' Pneumoconiosis

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Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Coal Workers' Pneumoconiosis (Black Lung Disease)	Coal dust **Carbon-laden macrophages**	Urban population Coal miners	Chronic bronchitis that progresses to progressive massive pulmonary fibrosis Complications: Caplan syndrome, Anthracosis	Fine nodular opacifications (**upper lungs**)

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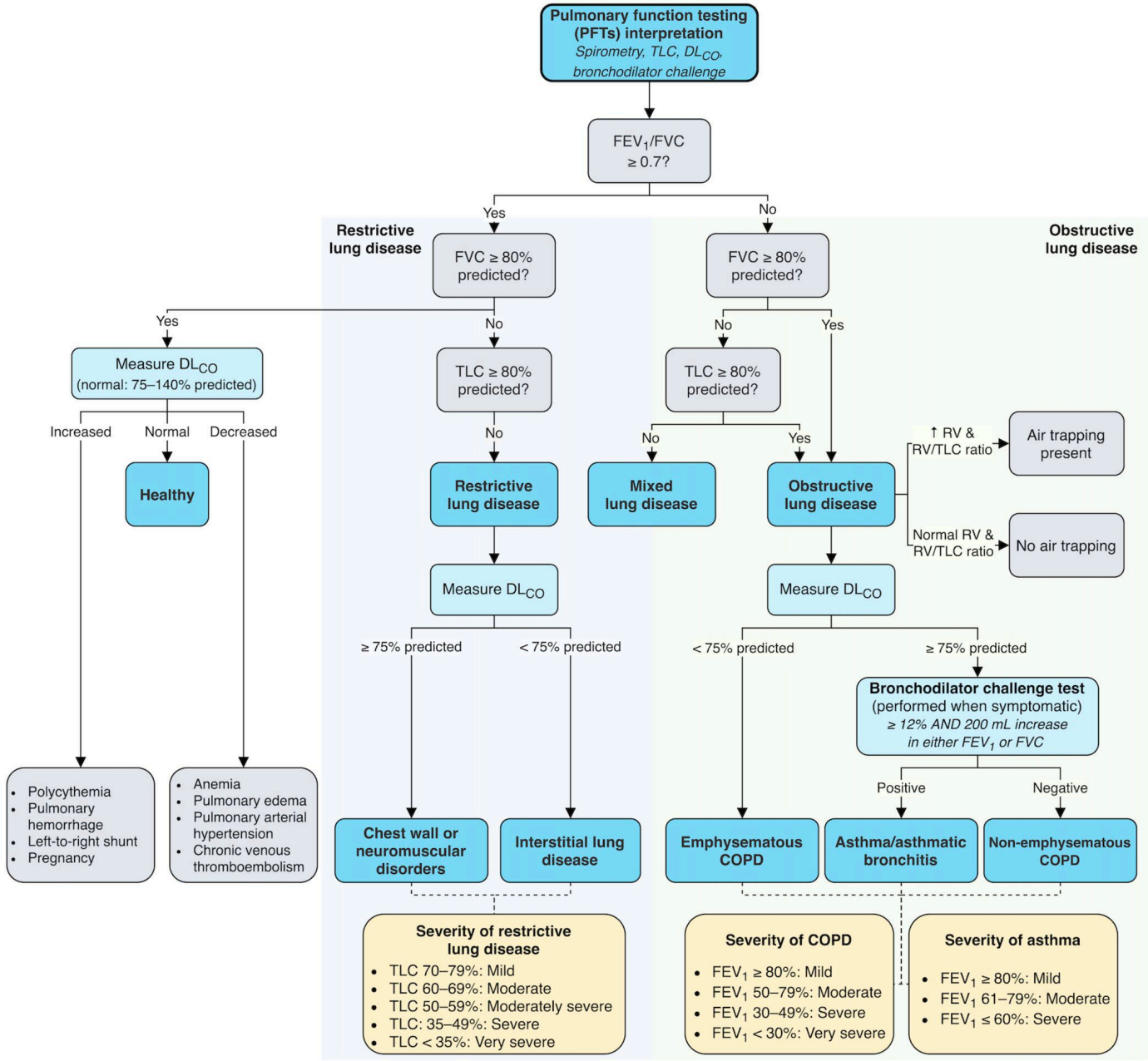
Pneumoconiosis: Berylliosis

Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Berylliosis	Beryllium	Aerospace engineering Ceramics industries Dental material production Nuclear and electronics plants Dye manufacturing Foundries	Progressive dyspnea within a few days of high-grade exposure **Noncaseating granulomatous** disease Complications: lung cancer, Cor pulmonale	Reticular interstitial pattern (**upper lobes**) +/- hilar lymphadenopathy

Summary

- Obstructive and restrictive lung diseases have different finding on spirometry and flow-volume loops that aid in diagnosis
- Symptomology and clinical presentations may overlap for obstructive and restrictive lung diseases; thus, it is important to know the diagnostic testing as well as findings
- The etiology and pathophysiology of obstructive and restrictive lung diseases can help explain symptomology
- Treatment modalities include pharmacological and non-pharmacological management and are important to be aware of
- Various, serious complications can arise from obstructive and restrictive lung diseases

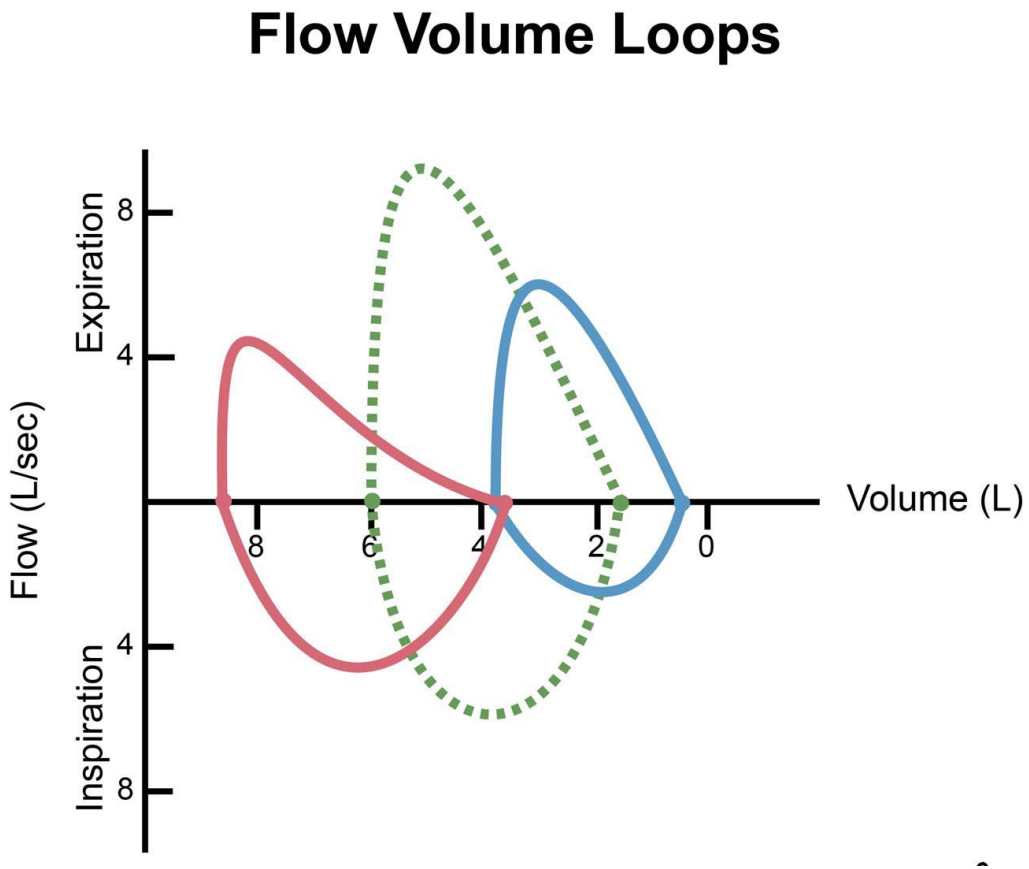


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Practice Question

A 56-year-old patient presents with dyspnea that has been occurring for 2 years getting progressively worse. On social history he endorses smoking 3 packs per day for 30 years. Chest x-ray reveals an increased anterior-posterior diameter with a flattened diaphragm. After stabilization with oxygen and pharmacological therapy he is referred to follow up at an outpatient pulmonologist. At his follow up, he undergoes spirometry testing.

Which color flow-volume loop would you expect to see in this patient?



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Practice Question

A 4-year-old boy is brought to the physician in December for episodic shortness of breath and a nonproductive cough for 3 months. These episodes frequently occur before sleeping, and he occasionally wakes up because of difficulty breathing. His mother also reports that he became short of breath while playing with his friends at daycare on several occasions. He is allergic to peanuts. He is at the 55th percentile for height and weight. Vital signs are within normal limits. Examination shows mild scattered wheezing in the thorax. An x-ray of the chest shows no abnormalities.

1. Which of the following is the most likely diagnosis?
   a) Cystic fibrosis
   b) Foreign body aspiration
   c) Asthma
   d) Chronic bronchitis
   e) Interstitial lung disease

2. What findings would most likely be seen on spirometry in this patient?
   a) Decreased FEV1
   b) Increased FEV1/FVC ratio
   c) Decreased TLC
   d) Decreased RV
   e) Normal FRC

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Practice Question

A 65-year-old woman with chronic obstructive pulmonary disease comes to the emergency department with a 2-day history of worsening shortness of breath and cough. She has a mild productive cough for the past 2 years but she noticed that her sputum is more yellow than usual over the past 2 days. She has not had any recent fevers, chills, sore throat, or a runny nose. Her only medication is a salmeterol inhaler that she uses twice daily. Her temperature is 36.7°C (98.0°F), pulse is 104/min, respirations are 28/min, and blood pressure is 134/73 mm Hg. Pulse oximetry shows an oxygen saturation of 85% on room air. She appears uncomfortable and shows labored breathing. Auscultation of the lungs discloses rhonchi that disappear after coughing and diffuse bilateral wheezes. A chest x-ray shows mild hyperinflation and flattening of the diaphragm but no consolidation. She is started on supplemental oxygen via nasal cannula.

Which of the following is the most appropriate initial pharmacotherapy?

a) Prednisone and roflumilast

b) Albuterol and montelukast

c) Prednisone and salmeterol

d) Albuterol and theophylline

e) Prednisone and albuterol

f) Prednisone and tiotropium

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Practice Question

A 33-year-old man is seen in the emergency department with pleuritic chest pain and shortness of breath. He has no history of serious illness and does not smoke but has worked for a pottery and glass manufacturing company for the last 10 years. An x-ray of the chest shows eggshell calcifications of the hilar lymph nodes and diffuse ground glass opacities in the upper lobes.

1. This patient is most likely to have which of the following diagnosis?
   a) Chronic bronchitis
   b) Berylliosis
   c) Silicosis
   d) Panacinar emphysema
   e) Asthma

2. This patient is at increased risk of developing which of the following long-term infectious complications?
   a) Mycobacterium avium complex
   b) Mycobacterium tuberculosis
   c) Pneumonia
   d) Aspergillosis
   e) Brucellosis

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Practice Question

A 27-year-old woman comes to the physician because of a 2-month history of increasing shortness of breath and a nonproductive cough. She has been unable to perform her daily activities. She has also had malaise and bilateral ankle pain during this period. She first noticed the symptoms after returning from a vacation to Arizona. She is a research assistant at the university's geology laboratory. She was born in Nigeria and visits her family there twice a year; her last trip was 3 months ago. Her temperature is 37.8°C (100.0°F), pulse is 100/min, respirations are 24/min, and blood pressure is 112/72 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. There is no palpable cervical or axillary lymphadenopathy. The lungs are clear to auscultation. Examination shows ciliary injection and photophobia of the left eye. The remainder of the examination shows no abnormalities. A complete blood count is within the reference range. An x-ray of the chest is shown.



1. Which of the following is the most likely diagnosis?
   a) Pulmonary tuberculosis
   b) Hypersensitivity pneumonitis
   c) Bronchiectasis
   d) Sarcoidosis
   e) Centrilobular emphysema

2. Which of the following additional findings is most likely to be found in this patient?
   a) Positive tuberculin skin test
   b) Low serum angiotensin-converting enzyme activity
   c) Elevated serum parathyroid hormone concentration
   d) Elevated CD4+ T-cell count in bronchoalveolar lavage
   e) Positive anti-dsDNA antibody testing